Assignment 2

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Q1. Online Food Delivery Application

food/model/FoodOrder.java

```
package food.model;

public abstract class FoodOrder {
    private int orderId;
    private String customerName;
    private double amount;
    private static String restaurantName = "Tasty Bites";
    private static int orderCount = 0;

    public FoodOrder(int orderId, String customerName, double amount) {
        this.orderId = orderId;
        this.customerName = customerName;
        this.amount = amount;
        orderCount++;
    }

    public int getOrderId() { return orderId; }
    public void setOrderId(int orderId) { this.orderId = orderId; }

    public String getCustomerName() { return customerName; }
    public void setCustomerName(String customerName) { this.customerName = customerName; }

    public double getAmount() { return amount; }
    public void setAmount(double amount) { this.amount = amount; }

    public static String getRestaurantName() { return restaurantName; }

    public abstract double calculateDeliveryCharge();

    public static void displayTotalOrders() {
        System.out.println("Total Orders: " + orderCount);
    }
}
```

food/model/Discountable.java

```
package food.model;

public interface Discountable {
    double applyDiscount();
}
```

food/model/RegularOrder.java

```
package food.model;

public class RegularOrder extends FoodOrder implements Discountable {

    public RegularOrder(int orderId, String customerName, double amount) {
        super(orderId, customerName, amount);
    }

    @Override
    public double calculateDeliveryCharge() {
        return 80.0;
    }

    @Override
```

```
    public double applyDiscount() {
        return getAmount() * 0.10;
    }
}
```

food/model/PremiumOrder.java

```
package food.model;

public class PremiumOrder extends FoodOrder implements Discountable {

    public PremiumOrder(int orderId, String customerName, double amount) {
        super(orderId, customerName, amount);
    }

    @Override
    public double calculateDeliveryCharge() {
        return 50.0;
    }

    @Override
    public double applyDiscount() {
        return getAmount() * 0.15;
    }
}
```

food/utility/OrderUtility.java

```
package food.utility;

public class OrderUtility {

    public static boolean validateAmount(double amount) {
        return amount > 0;
    }

    public static boolean validateCustomerName(String name) {
        return name != null && !name.trim().isEmpty();
    }

    public static String generateOrderSummary(int orderId, String customerName, double amount,
                                              double discount, double deliveryCharge, double finalAmount) {
        return "OrderID: " + orderId + ", Customer: " + customerName +
               ", Bill: " + amount + ", Discount: " + discount +
               ", Delivery: " + deliveryCharge + ", Final: " + finalAmount;
    }
}
```

food/main/MainApp.java

```
package food.main;

import food.model.*;
import food.utility.OrderUtility;

public class MainApp {
    public static void main(String[] args) {
        FoodOrder[] orders = new FoodOrder[6];
        orders[0] = new RegularOrder(1, "Aman", 500);
        orders[1] = new PremiumOrder(2, "Riya", 800);
        orders[2] = new RegularOrder(3, "Karan", 350);
        orders[3] = new PremiumOrder(4, "Simran", 1200);
        orders[4] = new RegularOrder(5, "Vikram", 600);
        orders[5] = new PremiumOrder(6, "Neha", 950);

        for (FoodOrder order : orders) {
            double deliveryCharge = order.calculateDeliveryCharge();
            double discount = ((Discountable) order).applyDiscount();
            double finalAmount = order.getAmount() - discount + deliveryCharge;
            if (OrderUtility.validateAmount(order.getAmount()) &&
```

```
                OrderUtility.validateCustomerName(order.getCustomerName())) {
                System.out.println(OrderUtility.generateOrderSummary(order.getOrderId(),
                        order.getCustomerName(), order.getAmount(), discount, deliveryCharge, finalAmount));
            }
        }
        FoodOrder.displayTotalOrders();
    }
}
```

Q2. Smart Library Resource Management System

library/model/LibraryResource.java

```
package library.model;

public abstract class LibraryResource {
    private int resourceId;
    private String title;
    private String author;
    private static String libraryName = "Central Library";
    private static int resourceCount = 0;

    public LibraryResource(int resourceId, String title, String author) {
        this.resourceId = resourceId;
        this.title = title;
        this.author = author;
        resourceCount++;
    }

    public int getResourceId() { return resourceId; }
    protected void setResourceId(int resourceId) { this.resourceId = resourceId; }

    public String getTitle() { return title; }
    public String getAuthor() { return author; }

    public static String getLibraryName() { return libraryName; }

    public abstract double calculateFine(int overdueDays);

    public static void displayTotalResources() {
        System.out.println("Total Resources: " + resourceCount);
    }
}
```

library/model/Printable.java

```
package library.model;

public interface Printable {
    void printDetails();
}
```

library/model/Book.java

```
package library.model;

public class Book extends LibraryResource implements Printable {

    public Book(int resourceId, String title, String author) {
        super(resourceId, title, author);
    }

    @Override
    public double calculateFine(int overdueDays) {
        return overdueDays * 5.0;
    }

    @Override
    public void printDetails() {
        System.out.println("Book -> ID: " + getResourceId() +
                ", Title: " + getTitle() + ", Author: " + getAuthor());
    }
}
```

library/model/DigitalResource.java

```
package library.model;
public class DigitalResource extends LibraryResource implements Printable {
```

```
    public DigitalResource(int resourceId, String title, String author) {
        super(resourceId, title, author);
    }

    @Override
    public double calculateFine(int overdueDays) {
        return overdueDays * 2.0;
    }

    @Override
    public void printDetails() {
        System.out.println("DigitalResource -> ID: " + getResourceId() +
                ", Title: " + getTitle() + ", Author: " + getAuthor());
    }
}
```

library/util/InputValidator.java

```
package library.util;

public class InputValidator {

    public static boolean validateResourceId(int id) {
        return id > 0;
    }

    public static boolean validateFineDays(int days) {
        return days >= 0;
    }
}
```

library/main/MainApp.java

```
package library.main;

import library.model.*;
import library.util.InputValidator;

public class MainApp {
    public static void main(String[] args) {
        LibraryResource[] resources = new LibraryResource[5];
        resources[0] = new Book(1, "Java Programming", "James Gosling");
        resources[1] = new DigitalResource(2, "Cloud Computing Basics", "Amit Sharma");
        resources[2] = new Book(3, "Data Structures", "Robert Lafore");
        resources[3] = new DigitalResource(4, "AI Fundamentals", "Andrew Ng");
        resources[4] = new Book(5, "Operating Systems", "Galvin");

        int[] overdueDays = {3, 5, 0, 7, 2};
        double totalFine = 0;

        for (int i = 0; i < resources.length; i++) {
            LibraryResource resource = resources[i];
            if (InputValidator.validateResourceId(resource.getResourceId()) &&
                InputValidator.validateFineDays(overdueDays[i])) {
                ((Printable) resource).printDetails();
                double fine = resource.calculateFine(overdueDays[i]);
                System.out.println("Overdue Days: " + overdueDays[i] + ", Fine: Rs. " + fine);
                totalFine += fine;
            }
        }
        System.out.println("Total Fine of all overdue resources: Rs. " + totalFine);
        LibraryResource.displayTotalResources();
    }
}
```